DO NOW — RECALL

# Lesson 091 · Fetch–Decode–Execute: A Register Trace

Name: _____ Class: _____ Date: _____

**1. Which register holds the current instruction?**

**2. What is the role of MAR?**

**3. Where is the next instruction address kept?**

**4. A full adder has A=1,B=0 and carry-in=1. State Sum and carry-out.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

CIR.

### 2. Model answer

Stores address being accessed.

### 3. Model answer

PC.

### 4. Model answer

1+0+1=$10_2$, so Sum0, carry-out1.

### By the end, you can…

• Fetch copies the address from PC to MAR; memory read transfers the instruction through MDR into CIR. In a sequential fetch, PC advances.

• The Control Unit decodes the current instruction held in CIR and produces the necessary control signals.

• Execution of LDA 80 reads the data held at memory address 80 into ACC; the next instruction is not executed until its own fetch.

### Vocabulary — know these terms

fetch — Retrieve next instruction.

decode — CU interprets instruction.

execute — Carry out operation.

CIR — Current Instruction Register.

ACC — Accumulator.

NEW LEARNING

# Read, annotate and explain

### Observe

Fetch copies the address from PC to MAR; memory read transfers the instruction through MDR into CIR. In a sequential fetch, PC advances.

### Explain

The Control Unit decodes the current instruction held in CIR and produces the necessary control signals.

### Evaluate

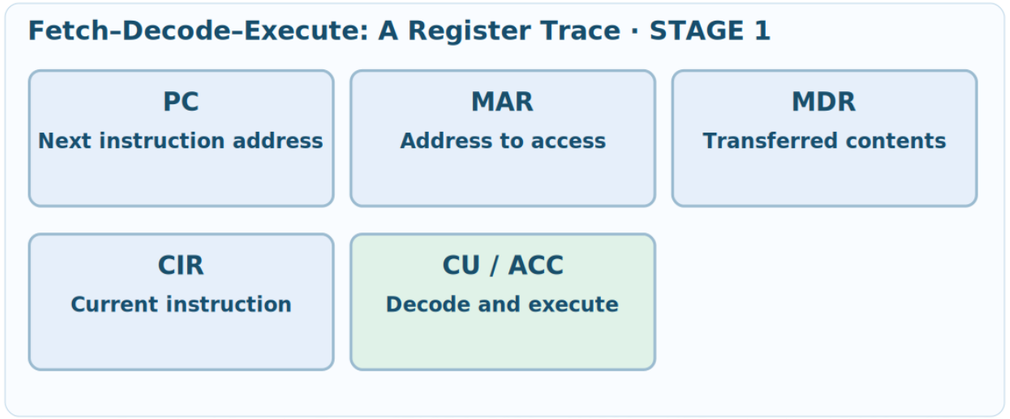
Execution of LDA 80 reads the data held at memory address 80 into ACC; the next instruction is not executed until its own fetch.

**Explain one step in your own words, using the correct technical terms.**

## New Learning · original lesson diagrams

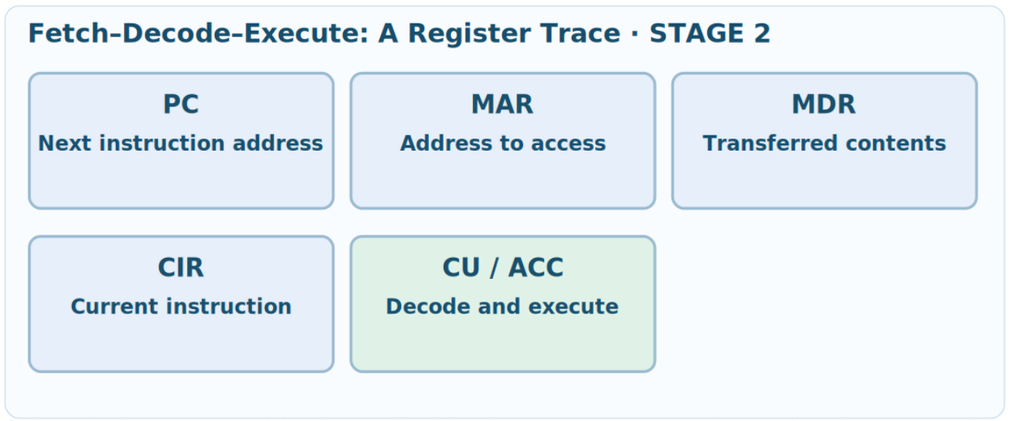
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Fetch–Decode–Execute: A Register Trace · STAGE 1; PC20 → MAR20 → MDR instruction LDA80 → CIR; PC21

**New Learning**




Fetch–Decode–Execute: A Register Trace · STAGE 2; PC20 → MAR20 → MDR instruction LDA80 → CIR; PC21; CU interprets LDA 80; identifies operand location



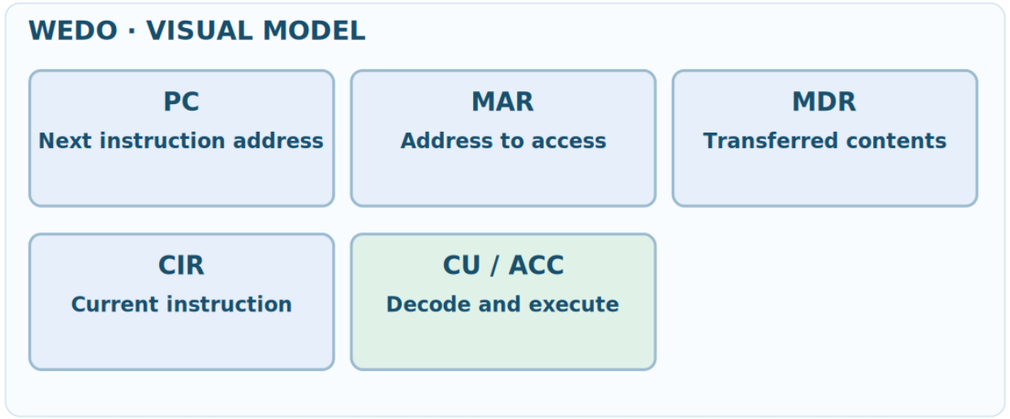


## We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**PC=42, memory[42]=SUB 90, memory[90]=3, ACC=11. Trace registers and result.**

After fetch PC43,CIR SUB90. Operand read from90 into MDR=3; ALU computes11−3, ACC8.




WEDO · VISUAL MODEL; After fetch PC43,CIR SUB90. Operand read from90 into MDR=3; ALU computes11−3, ACC8.

WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

### Guided 1 — CHECK

Explain the mechanism
After fetch PC43,CIR SUB90. Operand read from90 into MDR=3; ALU computes11−3, ACC8.
The addressed operand must not be confused with instruction address42.

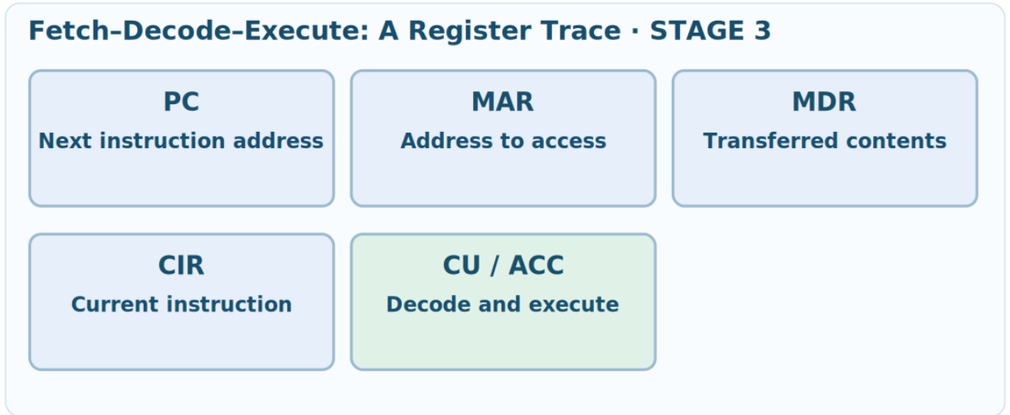




# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Fetch–Decode–Execute: A Register Trace · STAGE 3; PC20 → MAR20 → MDR instruction LDA80 → CIR; PC21; CU interprets LDA 80; identifies operand location; MAR80 → MDR13 → ACC13; next instruction PC21

I DO — WORKED EXAMPLE

# Follow the worked method

### PC=30, memory[30]=ADD 70, memory[70]=5, ACC=4. Trace after fetch and execute.

### Explain the mechanism

Fetch: MAR30, CIR ADD70, PC31. Execution reads memory[70]=5 and ALU adds to ACC4, giving ACC9.

The PC increments in fetch while the ALU operates on ACC and fetched operand.





EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**During a sequential fetch, MDR stores the instruction transferred from memory rather than its address.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

### GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 3

An answer states 'MDR stores address 80 and MAR stores the fetched 13'. Correct the error with a precise memory-read explanation. [6]



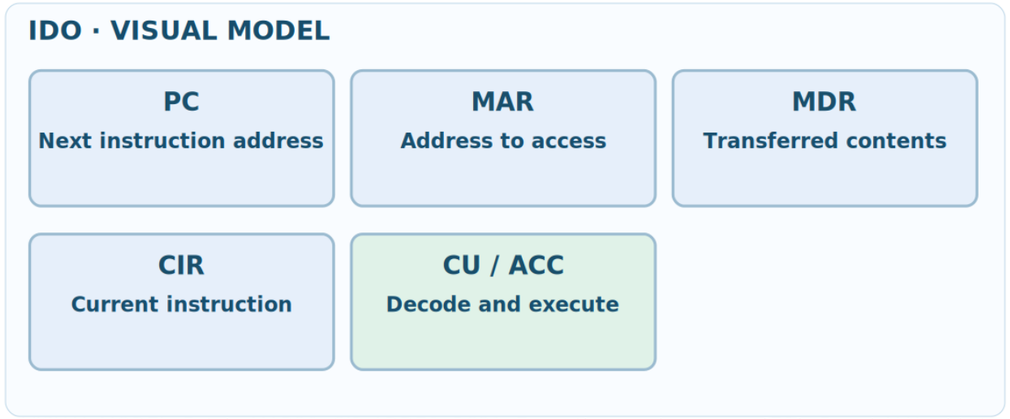


# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**PC=30, memory[30]=ADD 70, memory[70]=5, ACC=4. Trace after fetch and execute.**

Fetch: MAR30, CIR ADD70, PC31. Execution reads memory[70]=5 and ALU adds to ACC4, giving ACC9.




IDO · VISUAL MODEL; Fetch: MAR30, CIR ADD70, PC31. Execution reads memory[70]=5 and ALU adds to ACC4, giving ACC9.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

PC=42, memory[42]=SUB 90, memory[90]=3, ACC=11. Trace registers and result.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Trace the fetch of instruction at PC20 into MAR,MDR,CIR; give PC after sequential increment. [6]

### Question 2

Explain difference between decode and execute for LDA 80 when memory[80]=13. [6]